

Nadia Gul

Technical Writer | Machine Learning & Data Analysis

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OBJECTIVES

Computer Science graduate, MSCS, with solid expertise in technical writing, machine learning, and data analysis. Experienced in documenting ML pipelines, creating academic content, evaluating student work, and delivering insights through reports and visualizations. Skilled in communicating complex technical concepts clearly to diverse audiences.

RESEARCH EXPERIENCE

Teaching Assistant – Data Visualization

FAST-NUCES

Present
Islamabad, Pakistan

- Evaluated student assignments and projects using structured rubrics to ensure consistent and fair grading
- Designed assignments and coursework aligned with learning objectives and practical skills
- Provided feedback to students, helping them improve technical understanding and presentation of data
- Assisted students in understanding data visualization tools and concepts through clear explanations

Web Developer

Vantage Droid Private Limited

10/2023 – 02/2024

- Developed a student login system for authentication and user management
- Designed admin panel using SQL for database operations
- Built responsive UI using HTML, CSS, and JavaScript

EDUCATION

Master's in Computer Science

FAST-NUCES

2024 – 2026

CGPA: 3.19

Thesis (In Progress)

Resource-Aware Federated Learning for IoT: Efficient Client Selection and Lightweight Models

- Designing resource-aware client selection based on computation, energy, and communication constraints
- Developing a lightweight FL framework for heterogeneous IoT environments
- Integrating adaptive aggregation (synchronous, asynchronous, hybrid)
- Simulating non-IID data distributions
- Evaluating performance using accuracy, convergence, communication cost, and fairness metrics

Bachelor of Information Technology

University of Sargodha

• CGPA: 3.29

Badminton Ontology (Final Year Project)

- Developed an ontology-based semantic information retrieval system for the sports domain.
- Designed a structured framework to represent player skills, strategies, and equipment specifications.
- Implemented the Neon Methodology to construct an ontology network, facilitating knowledge reuse and collaborative development.
- Enhanced data management, analysis, and intelligent coaching system development through a standardized representation of badminton knowledge.

SKILLS

Technical Writing & Documentation

ML Reports, Research Writing, Instructional Content

LaTeX

Academic Writing, Report Formatting

Microsoft Office

Word, Excel, PowerPoint

Machine Learning

KNN, Decision Trees, Random Forest, Gradient Boosting

Data Analysis & Visualization

Tableau, Excel

Programming & Frameworks

- Python
- Scikit-learn

Data Science Fundamentals

- Data Preprocessing
- Feature Engineering

RESEARCH PROJECTS

Breast Cancer Prediction using K-Nearest Neighbors (KNN)

- Implemented KNN classification on the Breast Cancer Wisconsin dataset (569 samples, 30 features)
- Achieved 96.3% accuracy and 98% recall, emphasizing recall for medical diagnosis
- Performed preprocessing including feature scaling, correlation analysis, and outlier assessment
- Conducted sensitivity analysis on distance metrics and scaling techniques
- Applied PCA and feature pruning, reducing dimensionality by up to 40% with minimal performance loss
- Compared KNN and SVM models, analyzing trade-offs between accuracy and clinical reliability
- Documented methodology and results in a structured technical report using MS Word and LaTeX

Heart Disease Prediction using Decision Trees & Ensemble Methods

- Developed predictive models using Decision Trees, Random Forest, and Gradient Boosting
- Performed data preprocessing including missing value imputation, outlier removal, and feature encoding
- Addressed overfitting through pruning and hyperparameter tuning
- Achieved improved performance using ensemble methods (~81.7% accuracy with strong recall)
- Conducted bias-variance analysis and feature importance evaluation
- Proposed enhancements including SHAP-based explainability and probability calibration
- Prepared detailed analytical report with visualizations and evaluation insights

Systematic Literature Review on AI-based Clinical Trial Matching

- Conducted an in-depth analysis of AI techniques including NLP, ML, and hybrid approaches
- Evaluated datasets such as MIMIC-III, N2C2, and ClinicalTrials.gov
- Analyzed performance metrics including precision, recall, and F1-score
- Investigated EHR integration and AI-based patient-trial matching systems
- Documented findings in a structured academic research report

Interactive Data Storytelling with Tableau – FIFA Dataset

- Analyzed 17,000+ player dataset to uncover trends in ratings, age, and nationalities
- Built interactive dashboards with filters, KPIs, and visual storytelling features
- Cleaned and preprocessed data for accuracy and reliability

Disease Prediction System (Machine Learning)

- Developed predictive models using supervised learning algorithms
- Implemented Random Forest and Decision Tree classifiers with cross-validation
- Built an interface for inputting symptoms and generating predictions with confidence scores

VSAA-AC: Speaker-Invariant Accent Classification

Machine Learning | Speech Processing | PyTorch

- Developed a speaker-invariant English accent classification system using MFCCs, vowel formants (F1/F2), and Gradient Reversal Learning (GRL)
- Built and evaluated SVM, MLP, wav2vec2, and adversarial deep learning models on the Speech Accent Archive dataset

- Implemented audio preprocessing and feature extraction pipelines using Librosa and Parselmouth/Praat
- Applied speaker-held-out evaluation, ablation studies, confusion matrices, Macro-F1 analysis, and McNemar statistical testing
- Improved unseen-speaker generalization by reducing speaker identity leakage through adversarial training techniques

CERTIFICATIONS

- | | | |
|---|-----------------------------|---|
| • AWS Academy Graduate – Cloud Security Builder | • Introduction to AWS Cloud | • The Complete Web Development Bootcamp (Udemy) |
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LANGUAGES

- | | |
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| English | Urdu |
| Saraiki | Punjabi |

TECHNICAL WRITING EXPERIENCE

Machine Learning Technical Reports & Documentation

- Authored detailed technical reports on machine learning models including KNN, Decision Trees, and Ensemble Methods
- Documented complete ML pipelines: data preprocessing, feature engineering, model training, and evaluation
- Presented results using structured tables, charts, and evaluation metrics (Accuracy, F1-score, AUC-ROC)
- Applied academic writing standards using LaTeX and MS Word for professional documentation

Systematic Literature Review (AI in Healthcare)

- Conducted a comprehensive systematic literature review on AI-based clinical trial eligibility matching
- Synthesized research findings from multiple academic databases (IEEE, PubMed, Springer, Scopus)
- Structured research into methodology, datasets, evaluation metrics, and critical analysis
- Identified research gaps and proposed future directions in Explainable AI and healthcare systems

Teaching & Instructional Content (Data Visualization)

- Designed assignments and project tasks for students, ensuring clarity of instructions and learning objectives
- Created structured guidelines and evaluation rubrics to support fair and consistent grading
- Simplified complex data visualization concepts into step-by-step explanations for students

VOLUNTEER EXPERIENCE

Information Technology Society

University of
Sargodha